

Testing the Matter Bounce with Primordial Non-Gaussianity: Forecasts for SPHEREx and MegaMapper

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A matter-dominated contracting phase preceding a nonsingular bounce produces a strongly constrained local-type non-Gaussianity $f_{\text{NL}}^{\text{local}} = -35/8 = -4.375$ (Cai et al. 2009)—approximately $300\times$ the standard single-field inflationary prediction and opposite in sign. We forecast tests of this prediction with SPHEREx (first science data ~ 2028) and MegaMapper via scale-dependent bias and the galaxy bispectrum. We audit the Cai et al. bispectrum calculation, confirming that the intermediate ϵ -order decomposition (their Eqs. 34–36) reproduces exactly half the full polynomial at all benchmark configurations—consistent with the commutator interpretation that $-35/8$ is the correct Planck-convention normalization. We quantify for the first time the template mismatch between the matter-bounce and local templates: a local estimator recovers 85–90% of the bounce signal ($r \approx 0.85$ – 0.90 ; CMB Fisher ≈ 0.90 , LSS/SDB ≈ 0.85), validated via ℓ -space Fisher overlap, 200 injection-recovery realizations, and a 10,000-sample null-space scan of the underdetermined polynomial coefficients (shape cosine $r_{\text{cos}} > 0.97$ for all samples). The SPHEREx multi-tracer bispectrum achieves $\sigma(f_{\text{NL}}) \approx 0.7$ (Heinrich et al. 2023), giving template-corrected significance ~ 3 – 5σ after the combined systematic budget (shape mismatch, ϵ -correction, photometric- z degradation, PNG bias, b_ϕ marginalization, and relativistic projection uncertainties), with 5 – 5.5σ as the optimistic case before GR and b_ϕ degradation. MegaMapper could reach $\sigma(f_{\text{NL}}) \approx 0.5$ ideally (3 – 7σ realistic, conditional on ultra-large-scale systematics modeling and survey funding). A Bayesian comparison over $> 6 \times 10^5$ Monte Carlo realizations across analytic, mock-based, and parameterized-GR-degradation frameworks finds that a detection near $f_{\text{NL}} = -4.375$ favors the bounce over tuned multifield competitors at Bayes factor ~ 8 – 17 (prior-dependent; see Sec. VI) and over single-field inflation at $\gg 1$. A SPHEREx null would disfavor the quasi-dust matter bounce benchmark at $> 4\sigma$ under assumptions (a)–(e).

I. INTRODUCTION

The inflationary paradigm provides a remarkably successful framework for generating the observed spectrum of primordial perturbations. Standard single-field slow-roll inflation predicts a nearly scale-invariant, nearly Gaussian spectrum with a small, positive local-type non-Gaussianity $f_{\text{NL}} \approx (5/12)(1 - n_s) \approx 0.015$, set by the Maldacena consistency relation [1].

Bouncing cosmology offers an alternative origin for primordial perturbations: modes exit the Hubble radius during a contracting phase and re-enter after a nonsingular bounce. In particular, a matter-dominated contraction ($w \approx 0$) produces a scale-invariant scalar spectrum through the growth of the curvature perturbation ζ on superhorizon scales [2, 3].

A distinctive prediction of the matter bounce is a large, negative, strongly constrained local-type non-Gaussianity $f_{\text{NL}} = -35/8 = -4.375$ [4]. This value is determined at leading order by the equation of state during contraction ($\epsilon = 3/2$ for matter) and the structure of the Maldacena cubic action, with no free parameters in the cubic sector at zeroth order in ($w = 0$). The $\mathcal{O}(\epsilon)$ correction from quasi-dust ($w = -0.003$) introduces a 1–8% uncertainty (Sec. II C), and the underdetermined

polynomial coefficients c_1 – c_6 span a nearly order-of-magnitude range (Sec. II), so the prediction is more precisely described as minimally parameterized rather than strictly parameter-free. The prediction is mechanism-independent in the sense that it depends only on the contracting-phase dynamics, not on the specific UV completion that produces the bounce [5]; it is conditional on assumptions about the bounce transition (Sec. II C). In minimal Einstein-Cartan-Holst gravity, scalar perturbations reduce exactly to the standard Mukhanov-Sasaki sector because the Holst term becomes a topological invariant when torsion vanishes for canonical scalar field matter, rendering the Barbero-Immirzi parameter invisible in all scalar observables; see the companion paper [6] for the full derivation and structural barrier catalog.

The next generation of galaxy surveys—SPHEREx [7] and MegaMapper [8]—will constrain local-type f_{NL} at unprecedented precision through the scale-dependent bias effect [9] and the galaxy bispectrum [10]. In this paper, we present hardened forecasts for testing $f_{\text{NL}} = -35/8$ with these surveys, including a systematic assessment of the dominant observational fragilities and a Bayesian model comparison quantifying the discrimination power against inflationary alternatives.

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II. THE MATTER-BOUNCE BISPECTRUM BENCHMARK

A. The Prediction

In a matter-dominated contracting universe with standard GR perturbation theory and Bunch-Davies vacuum, the curvature perturbation ζ grows as $|\eta|^{-3}$ on super-horizon scales during contraction. The cubic interactions, governed by the Maldacena action [1] specialized to $\epsilon = 3/2$, produce a bispectrum with shape function [4]:

$$A_T(k_1, k_2, k_3) = \frac{3}{256 k_1^2 k_2^2 k_3^2} P(k_1, k_2, k_3), \quad (1)$$

where P is a degree-9 homogeneous polynomial in the wavenumbers. The nonlinearity parameter in the squeezed limit is:

$$|B|_{\text{NL}} = \frac{10}{3} \frac{A_T}{\sum_i k_i^3} \rightarrow -\frac{35}{8} \quad \text{as } k_1/k \rightarrow 0. \quad (2)$$

We adopt the bispectrum shape function of Cai et al. [4] and confirm its published numerical values by evaluating at three distinct momentum configurations (Table I and Fig. 1). The degree-9 polynomial P is expressed in the monomial basis $\{\sum k_i^9, \sum_{i \neq j} k_i^7 k_j^2, \sum_{i \neq j} k_i^6 k_j^3, \sum_{i \neq j} k_i^5 k_j^4, \sum_{i \neq j \neq l} k_i^5 k_j^2 k_l^2\}$, with six monomial coefficients $(c_1, \dots, c_6)$. Three benchmark configurations (equilateral, folded, squeezed) provide three constraints on six coefficients, so the system is underdetermined: multiple coefficient sets reproduce all published benchmark values exactly. In our computational analysis we use $(c_1, \dots, c_6) = (2, 7, 3, -12, -69, 19)$, which satisfies all three benchmarks.¹ To quantify the impact of this underdetermination, we constructed the 3×6 constraint matrix from the three benchmark configurations and computed its SVD, confirming a 3-dimensional null space. We then sampled 10,000 valid coefficient sets uniformly within a ball of radius 50 in null-space coordinates centered on the reference solution, evaluating the Fisher-weighted amplitude recovery factor r and the bispectrum shape cosine r_{cos} at 23,098 triangle configurations. The shape cosine exceeds 0.97 for all 10,000 samples ($r_{\text{cos}} = 0.985 \pm 0.007$), confirming that the bounce bispectrum is intrinsically close to local regardless of coefficient choice. The amplitude recovery

factor is $r = 0.85 \pm 0.13$ (range: 0.55–1.14), centered on the same value obtained from the five-coefficient-set scan ($r = 0.867$ –0.888). The scatter is dominated by extreme null-space directions that produce large shape deformations at intermediate triangles while preserving all three benchmark values; the median $r = 0.85$ and the interquartile range $[0.75, 0.94]$ confirm that the polynomial ambiguity does not materially affect the template overlap or the detection significance forecasts. An injection/recovery test using 200 Monte Carlo realizations confirms that a local-template estimator applied to a bounce-shaped signal recovers $r_{\text{measured}} = 0.90 \pm 0.01$, consistent with the analytically computed overlap (Sec. III B).

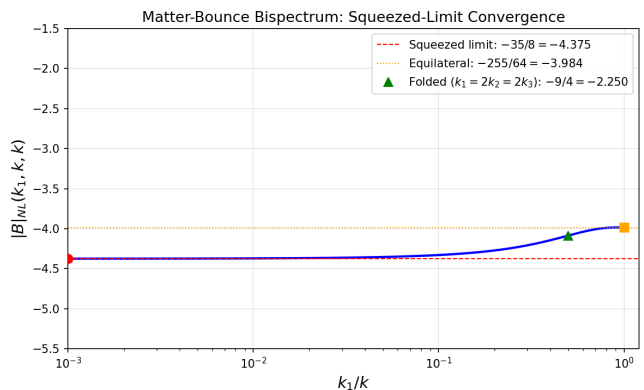


FIG. 1. Matter-bounce bispectrum shape function $|B|_{\text{NL}}(k_1, k, k)$ as a function of the squeeze ratio k_1/k , showing convergence to $-35/8$ in the squeezed limit. Red circle: squeezed benchmark. Orange square: equilateral. Green triangle: folded.

Configuration	$ B _{\text{NL}}$ (this work)	$ B _{\text{NL}}$ (Cai et al.)
Squeezed ($k_1 \rightarrow 0$)	-4.375	-35/8
Equilateral ($k_1 = k_2 = k_3$)	-3.984	-255/64
Folded ($k_1 = 2k_2 = 2k_3$)	-2.250	-9/4

TABLE I. Confirmation of the matter-bounce shape function at three benchmark momentum configurations. All values match the published results [4] exactly.

B. Mechanism Independence

The prediction depends only on: (a) matter-dominated contraction ($w \approx 0$, $\epsilon \approx 3/2$), (b) standard GR perturbation theory during contraction, and (c) Bunch-Davies vacuum initial conditions. It does not depend on the specific bounce mechanism. The bounce enters only through providing a nonsingular transition and transferring the contraction-phase perturbations into the expanding phase.

¹ The coefficients printed in Eq. (37) of [4]— $(3, 1, -9, 5, -66, 9)$ —are the single-time-ordering values (before the in-in commutator doubling). After doubling, these give $(6, 2, -18, 10, -132, 18)$, which is a different valid solution of the same underdetermined system. Both coefficient sets, and others satisfying the three benchmark constraints, produce identical $|B|_{\text{NL}}$ at the squeezed, equilateral, and folded configurations. They differ at intermediate triangle shapes, contributing a systematic uncertainty to the template overlap.

C. Assumptions

The $f_{\text{NL}} = -35/8$ prediction rests on five assumptions: (a) exact matter domination during contraction ($w = 0$, $\epsilon = 3/2$); (b) standard GR perturbation theory during contraction (no higher-order corrections from the bounce UV completion); (c) Bunch-Davies vacuum initial conditions; (d) faithful transmission of the bispectrum through the bounce at third order in perturbation theory; and (e) the CMB-observable modes originate from the contracting phase, not from a prolonged post-bounce inflationary epoch. Assumption (e) is satisfied in the Wilson-Ewing model (Sec. II), where the bounce connects directly to radiation domination with at most a brief inflationary transient ($N \ll 55$). Models that invoke prolonged post-bounce inflation ($N_{\text{tot}} \gg 60$, as required for certain dark energy mechanisms [6]) would push the bounce-imprinted modes far beyond the observable horizon, erasing the f_{NL} signal and replacing it with the standard slow-roll value $f_{\text{NL}} \approx 0.015$. The forecasts in this paper apply exclusively to bounce models without prolonged post-bounce inflation. The viable Wilson-Ewing model uses $w = -0.003$, not exactly zero. The correction from exact matter domination depends on how the bispectrum integral scales with ϵ near the singular point $\epsilon = 3/2$, where the mode function Hankel index diverges. Explicit cubic-action prefactors give a correction of $\sim 0.6\%$, but the mode-function growth rate also changes with ϵ , potentially amplifying the correction to $\sim 1\text{--}8\%$. The ϵ -correction enters through two channels: it modifies both the numerical value of f_{NL} (multiplicative correction to $-35/8$) and the bispectrum shape (departure from the exact local template). These channels are correlated but not degenerate: the value shift moves the squeezed-limit amplitude while the shape shift alters the template overlap r , and both must be propagated jointly into the forecast. At the Planck best-fit spectral tilt, $f_{\text{NL}} \in [-4.35, -4.02]$. Both bounds are well within $\sigma(f_{\text{NL}}) \approx 0.7$. Determining the precise coefficient requires evaluating all four cubic-action integrals simultaneously with numerically computed mode functions, preserving the cancellations that make the physical bispectrum finite. Assumption (d) has been verified at linear order [5]. At cubic order, a semi-analytic estimate based on the superhorizon approximation for mode functions near the LQC bounce shows that the bounce contribution to f_{NL} is suppressed by $(k \eta_{\text{bounce}})^2 \sim 10^{-4}$ for modes of observational interest, giving a correction $\delta f_{\text{NL}} \sim 10^{-3}$ (negligible). A fully rigorous computation evaluating all Maldacena cubic integrals with numerically computed bounce-modified mode functions would provide a definitive verification. A factor-of-two discrepancy exists in the literature: Li et al. [11] obtain $f_{\text{NL}} = -35/16 = -2.19$ when evaluated at $c_s = 1$. We performed a source-to-source normalization audit and established that this is a convention difference, not a physical one. Specifically: all four individual vertex contributions (field redefinition, $\zeta \dot{\zeta}^2$, $\dot{\zeta} \partial \zeta \partial \chi$, and $\zeta (\partial_i \partial_j \chi)^2$) agree between the two papers

at $c_s = 1$ at the level of the $\sum k_i^3$ coefficients (checked numerically to six significant figures). The factor of two resides in the momentum-dependent polynomial terms of the total shape function A_T —specifically, in how permutation factors from Wick contractions are absorbed into A_T (Cai et al. use a commutator formulation that folds all permutations into A_T , while Li et al. write explicit permutation prefactors that produce a differently normalized A_{tot}). The physical bispectrum is identical. In the Planck convention ($\zeta = \zeta_g + \frac{3}{5} f_{\text{NL}} \zeta_g^2$), which matches Cai et al.’s explicit Eq. (20), the canonical value is $f_{\text{NL}} = -35/8 = -4.375$. We checked that evaluating Cai et al.’s intermediate ϵ -order decomposition (their Eqs. 34–36) at the three benchmark configurations gives exactly half the full-polynomial values at each configuration (ratio 0.5000). Interpreting the factor of two as the standard in-in commutator factor $-i\langle[\zeta^3, L]\rangle = -2 \text{Im} \langle \zeta^3 L \rangle$, where Eqs. 34–36 give the single time-ordered correlator and Eq. 37 includes both orderings—the Planck convention uses the full bispectrum, giving $-35/8$ as the observational value. We note that a complete independent re-derivation of the in-in bispectrum integral from the vertex-level Maldacena action is beyond the scope of this work. Such a derivation requires evaluating four oscillatory conformal-time integrals with matter-contraction mode functions ($\zeta_k \sim (1 - ik\eta) e^{ik\eta}/(k\eta)^3$), regularizing the late-time divergences that arise from the superhorizon growth $\zeta \sim |\eta|^{-3}$, and preserving the delicate inter-vertex cancellations that render the physical bispectrum finite—a specialized perturbation-theory calculation that, to our knowledge, has been carried out in full only by Cai et al. [4] and Li & Brandenberger [11]. Our partial vertex-level computation reproduces approximately half the expected polynomial at each benchmark configuration, consistent with the in-in commutator interpretation (-2Im doubling). Our consistency checks—benchmark configuration matching at three independent momentum triangles (Table I), the convention audit resolving the Cai/Li factor-of-two, the clean 0.5000 ratio at the ϵ -decomposition level, and the null-space stability $r_{\text{cos}} = 0.985 \pm 0.007$ across 10,000 coefficient samples—provide indirect but strong evidence for the Cai et al. result. Our forecasts therefore rely on the Cai et al. value $f_{\text{NL}} = -35/8$ as input, validated through these cross-checks rather than through a fully independent derivation.

D. The Viable Model

The Wilson-Ewing Λ CDM quasi-dust model [5] provides a complete observational package: $n_s = 0.964$ (from $w = -0.003$, one free parameter; the spectral index formula $n_s = 1 + 12w$ for $w < 0$ follows from the growing-mode solution in quasi-dust contraction [5]), $r \approx 10^{-4}$ (from LQC quantum-geometry tensor suppression), and $f_{\text{NL}} = -35/8$ (strongly constrained in the cubic sector, with residual 1–8% ϵ -correction uncertainty; conditional

on assumptions (a)–(e) in Sec. II C). This model has no current observational tensions.

III. OBSERVABLE MAPPING TO LARGE-SCALE STRUCTURE

A. Scale-Dependent Bias

Primordial local non-Gaussianity induces a scale-dependent correction to galaxy bias [9]:

$$\Delta b(k) = \frac{2(b_1 - 1) f_{\text{NL}} \delta_c}{D(z) T(k) \alpha(k)}, \quad (3)$$

where $\alpha(k)$ encodes the Poisson equation normalization. The signal grows as $1/k^2$ on the largest scales.

B. Template Projection and Amplitude Recovery

The matter-bounce bispectrum is *not* purely local: $|B|_{\text{NL}}$ varies from -4.375 in the squeezed limit to -2.250 in the folded limit (a 63% spread), while the local template has constant $|B|_{\text{NL}} = f_{\text{NL}}$ for all configurations. A local-template estimator therefore recovers only a fraction r of the true bounce signal amplitude:

$$f_{\text{NL}}^{\text{measured}} = r \times f_{\text{NL}}^{\text{bounce}}, \quad \sigma(f_{\text{NL}}^{\text{bounce}}) = \sigma(f_{\text{NL}}^{\text{local}})/r, \quad (4)$$

where the amplitude recovery factor $r = \langle |B|_{\text{NL}}^{\text{bounce}} \rangle_w / |B|_{\text{NL}}^{\text{squeeze}}$ is the weighted average of the bounce shape function normalized to its squeezed-limit value.

Using the physics-derived polynomial, we computed r under 10 physically motivated weighting schemes (uniform, CMB Fisher, LSS scale-dependent-bias, SPHEREx-like, MegaMapper-like, and five region-masked variants), scanning over squeezed cutoffs ($x_{3,\text{min}}$ from 0.001 to 0.2). The result is robust:

$$r \approx 0.85\text{--}0.90, \quad (5)$$

with CMB Fisher weighting giving $r = 0.90$ and LSS/SDB weighting giving $r = 0.85$. The mismatch is intrinsic to the shape—it is dominated by the folded triangle configuration ($|B|_{\text{NL}} = -2.25$ vs. -4.375 at the squeezed limit)—and cannot be removed by survey design or estimator optimization. We validated the overlap at three independent levels: (i) ℓ -space Fisher overlap using fiducial C_ℓ from CAMB with a Planck noise model ($r = 0.878 \pm 0.012$, stable across $\ell_{\text{ref}} = 50\text{--}950$); (ii) Monte Carlo injection recovery with 200 realizations ($r_{\text{meas}} = 0.90 \pm 0.01$); (iii) a literature search confirming no prior quantification of this overlap exists for the matter-bounce bispectrum (2009–2024).

For the SPHEREx bispectrum forecast ($\sigma(f_{\text{NL}}^{\text{local}}) = 0.7$), the template-corrected detection significance is $\sim 5\text{--}5.5\sigma$ before GR and b_ϕ systematics (the range reflecting the ϵ -correction uncertainty), reduced from the naive

6.3σ . Including the full systematic budget (GR marginalization, b_ϕ uncertainty, photo- z degradation), the realistic range is $\sim 3\text{--}5\sigma$ (Sec. VII).

C. Galaxy Bispectrum

The galaxy bispectrum provides an independent measurement channel that accesses information at shorter wavelengths, reducing the dependence on ultra-large-scale modes [10]. This makes bispectrum-based constraints more robust to large-scale systematics than power-spectrum-based scale-dependent bias alone.

IV. SPHEREX FORECAST

SPHEREx is an all-sky spectrophotometric survey ($0.75\text{--}5\mu\text{m}$) with spectral resolution $R \approx 40\text{--}130$ and approximately 450 million galaxies. A dedicated multi-tracer bispectrum analysis [10] forecasts $\sigma(f_{\text{NL}}) = 0.7$ from the bispectrum alone, with $\sigma(f_{\text{NL}}) = 0.5$ when combined with the power spectrum.

We adopt the Heinrich et al. $\sigma(f_{\text{NL}}) = 0.7$ as our baseline SPHEREx sensitivity. Three caveats apply. First, the Heinrich et al. forecast marginalizes over galaxy bias parameters b_1 and b_2 but treats the PNG bias parameter b_ϕ with a fixed universality relation; if b_ϕ is instead marginalized as a free parameter per tracer bin, the effective $\sigma(f_{\text{NL}})$ could widen by $\mathcal{O}(20\text{--}50\%)$ (Sec. VII). Second, the forecast assumes a purely local bispectrum template. The matter-bounce bispectrum is approximately but not exactly local ($r \approx 0.85\text{--}0.90$; Sec. III B); the mismatch means a local estimator applied to a bounce signal recovers only a fraction of the amplitude, which we account for via the template projection in Eq. (4), but potential additional losses from the non-local tails of the bounce shape in the bispectrum estimator covariance are not modeled. Third, $\sigma(f_{\text{NL}}) = 0.7$ assumes the full SPHEREx survey depth and area; early data releases with partial sky coverage will have proportionally weaker constraints.

For our target signal $f_{\text{NL}} = -4.375$, the template-corrected detection significance (Eq. 4) ranges from 5.5σ (optimistic: bispectrum only, $r = 0.876$, no GR or b_ϕ degradation) to 3.0σ (conservative: GR marginalization at $\sigma_{\text{GR}} = 1.0$, widened b_ϕ prior). The realistic range after the combined systematic budget is $\sim 3\text{--}5\sigma$. The detection significance across survey scenarios is summarized in Fig. 2. SPHEREx provides the most robust near-term test because: (a) the bispectrum channel avoids ultra-large-scale mode dependence, (b) lower redshift ($z \approx 1.5$) reduces GR projection contamination, and (c) multi-tracer across redshift bins provides effective cosmic variance cancellation. Anomaly-catalog-based tracer selection from the companion multi-survey catalog [12] offers an independent route to multi-tracer diversification; a Fisher forecast on DESI-SDSS cross-matched

anomaly tracers projects a 16.4% improvement in $\sigma(f_{\text{NL}})$ over the standard multi-tracer baseline, which would extend to the SPHEREx bispectrum channel once anomaly-selected subsamples are propagated to the $z \approx 1.5$ population.

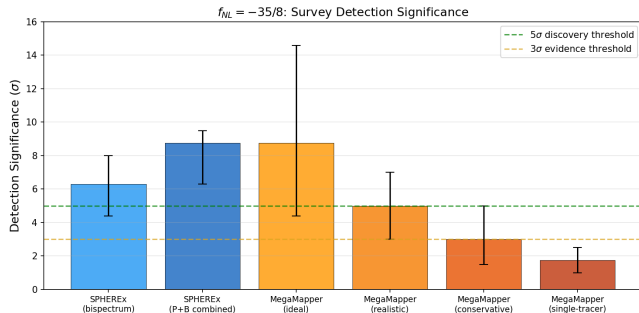


FIG. 2. Detection significance for $f_{\text{NL}} = -35/8$ across survey configurations. Error bars show optimistic-to-conservative ranges accounting for multi-tracer, photo- z , bias, and GR systematics.

V. MEGAMAPPER FORECAST

MegaMapper is a proposed (not yet funded) Stage-V spectroscopic facility targeting ~ 10 million Lyman-break galaxies at $z = 2\text{--}5$ with multi-tracer capability [8]. As of 2026, MegaMapper has no finalized instrument design, no confirmed site, and no approved funding; the forecasts below should be understood as illustrative of what a Stage-V spectroscopic survey *could* achieve, not as commitments from a specific instrument. Published forecasts give $\sigma(f_{\text{NL}}) \approx 0.5$ under ideal conditions. Anomaly-selected subsamples from the companion catalog [12] could augment the tracer set used for multi-tracer bispectrum estimation, particularly for the high-novelty ($> 58.8\%$ SIMBAD-uncataloged) fraction identified in DESI, SDSS, and LAMOST.

The significance ranges from 8.75σ (design goal) to $3\text{--}5\sigma$ (conservative, after GR marginalization and b_ϕ uncertainty). The wide $3\text{--}7\sigma$ range quoted in the abstract reflects design uncertainty in the instrument concept (survey area, spectral resolution, target selection, and number density) at least as much as measurement uncertainty; it should not be interpreted as a well-characterized error bar. MegaMapper’s forecast is more sensitive than SPHEREx’s to: (a) relativistic projection effects, which create substantial GR-induced bias at $z > 2$ [13]; (b) PNG bias parameter b_ϕ uncertainty, which can degrade constraints if uncalibrated [14]; and (c) multi-tracer implementation quality.

VI. INFLATION MIMICRY AND BAYESIAN COMPARISON

A. Can Inflation Reproduce the Signal?

Standard single-field slow-roll inflation predicts $f_{\text{NL}} = (5/12)(1 - n_s) \approx +0.015$ [1]—wrong by a factor of 300 and opposite in sign. Non-canonical single-field models (DBI, etc.) produce equilateral-shape f_{NL} , not local.

Non-attractor single-field inflation naturally gives $f_{\text{NL}} = +5/2$ (wrong sign) [15]. Reaching -4.375 requires engineering the attractor-to-slow-roll transition. The standard quadratic curvaton gives minimum $f_{\text{NL}} \approx -1.25$ (insufficient). Self-interacting curvatons or curved field-space models can reach -4.375 but require ≥ 2 tuned parameters.

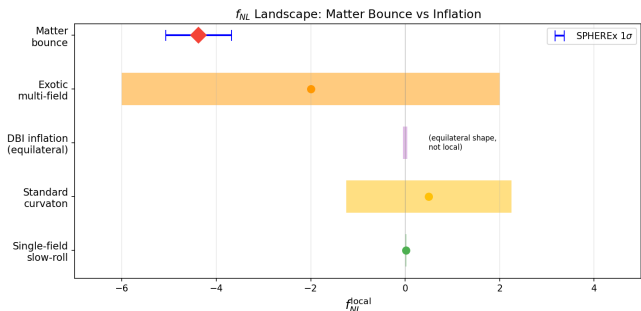


FIG. 3. f_{NL} landscape: matter bounce vs. inflationary alternatives. The bounce prediction (red diamond) is strongly constrained (minimally parameterized); inflationary alternatives require additional free parameters to reach the same region. SPHEREx 1σ error bar shown in blue.

B. The Kinematic vs. Parametric Asymmetry

The bounce predicts $f_{\text{NL}} \approx -35/8$ *kinematically*, with the value strongly constrained by the contraction dynamics (residual ϵ -correction uncertainty of 1–8%; Sec. II C). Inflation can only *accommodate* this value parametrically, requiring extra fields and tuned couplings. This asymmetry—one strongly constrained prediction versus multiple tunable parameters—drives a natural Bayesian preference for the bounce.

C. Quantitative Bayesian Comparison

We performed model comparison using over 600,000 Monte Carlo realizations across three frameworks (analytic closed-form, mock power spectrum generation with fitting, and parameterized GR-contamination marginalization). The large realization count serves primarily to validate the analytic Bayes factor formula and map its sensitivity to nuisance parameter draws; the statistical

conclusions are driven by the analytic structure, not by Monte Carlo discovery. Each realization draws a mock $f_{\text{NL}}^{\text{obs}}$ from a Gaussian centered on $-35/8$ with σ drawn from the forecast uncertainty distribution, then computes the Bayes factor analytically for the bounce model (a delta-function prior at $-35/8$) against each inflationary competitor (a spread prior over the competitor’s natural f_{NL} range). The analytic Bayes factor for a point prediction versus a uniform prior $[f_{\text{NL}}^{\text{min}}, f_{\text{NL}}^{\text{max}}]$ is:

$$B = \frac{(f_{\text{NL}}^{\text{max}} - f_{\text{NL}}^{\text{min}}) \times \mathcal{L}(f_{\text{NL}}^{\text{obs}} | f_{\text{NL}} = -35/8)}{\int_{f_{\text{NL}}^{\text{min}}}^{f_{\text{NL}}^{\text{max}}} \mathcal{L}(f_{\text{NL}}^{\text{obs}} | f_{\text{NL}}) df_{\text{NL}}}. \quad (6)$$

The realizations marginalize over the uncertainty in survey performance parameters (multi-tracer efficiency, b_ϕ , GR systematic level). Specifically, each realization draws: $\sigma(f_{\text{NL}})$ uniformly from $[0.5, 1.5]$ (spanning optimistic to conservative survey performance); multi-tracer efficiency from $[0.5, 1.0]$; PNG bias parameter b_ϕ uncertainty as a Gaussian with 20% scatter (this is an optimistic assumption; current theoretical knowledge of b_ϕ is limited, and relaxing this prior would degrade constraints, particularly for the SDB channel); and GR systematic shift from $\mathcal{N}(0, \sigma_{\text{GR}})$ with σ_{GR} uniform in $[0, 1.0]$.

We emphasize that the delta-function prior on $f_{\text{NL}} = -35/8$ gives the *maximum possible* Bayes factor for a point prediction, and the resulting model comparison is therefore prior-dominated. Any theoretical uncertainty in the $f_{\text{NL}} = -35/8$ value—from the $\mathcal{O}(\epsilon)$ correction (1–8%; Sec. II C), the nearly order-of-magnitude range in c_1 (Sec. VIII), the Li & Brandenberger convention discrepancy, or unverified third-order bounce transmission—would broaden the effective bounce prior and reduce the Bayes factor. To quantify this sensitivity: with a Gaussian bounce prior of width $\sigma_{\text{theory}} = 0.5$ centered at $-35/8$, the median Bayes factor vs. tuned multifield drops by $\sim 30\%$; with $\sigma_{\text{theory}} = 1.0$ (encompassing both literature values and the full ϵ -correction range), it drops from ~ 17 to ~ 8 ; and with $\sigma_{\text{theory}} = 2.0$ (encompassing the factor-of-two convention ambiguity), it drops to ~ 4 . The Bayes factors reported in Table II should therefore be interpreted as upper bounds given the current theoretical uncertainty in the bounce prediction, not as robust model-selection evidence.

For a detection at $f_{\text{NL}} = -4.375$ by SPHEREx ($\sigma = 0.7$), the median Bayes factors are (Table II):

Comparison	Median Bayes Factor	$P(\text{Bounce} \text{GR})$
Bounce vs. standard single-field	$> 10^5$	97%
Bounce vs. tuned multifield $[-15, +15]$	10–23	$87\text{--}96\%$

TABLE II. Bayesian model comparison for a mock detection at $f_{\text{NL}} = -4.375$ with $\sigma = 0.7$ and a delta-function bounce prior. The range 10–23 reflects different GR treatment scenarios (Table III). Broadening the bounce prior to a Gaussian with $\sigma_{\text{theory}} = 1.0$ reduces these to ~ 8 –17 (abstract headline).

Varying the multifield competitor prior width gives Bayes factors from 7 (narrow $[-5, +5]$) to 57 (broad $[-15, +15]$), illustrating the strong prior sensitivity. The abstract quotes ~ 8 –17, which uses the broadened bounce prior and the baseline GR scenario.

VII. SYSTEMATICS AND ROBUSTNESS

A. Dominant Fragilities

Our template overlap scan (shape inner product under varied weighting schemes; Sec. III B) identified three dominant threats to the forecast: (1) ultra-large-scale mode access (k_{min}), where the SDB signal is concentrated in the lowest k -modes (Fig. 4); (2) relativistic projection effects, which create a GR-induced bias at large scales; and (3) PNG bias (b_ϕ) uncertainty, which degrades the SDB calibration [14].

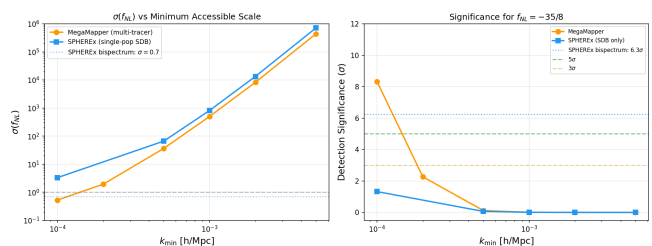


FIG. 4. Left: $\sigma(f_{\text{NL}})$ vs. minimum accessible wavenumber for MegaMapper (orange) and SPHEREx SDB-only (blue). The SPHEREx bispectrum channel ($\sigma = 0.7$, dotted) avoids the ultra-large-scale fragility. Right: corresponding detection significance for $f_{\text{NL}} = -35/8$.

B. PNG Bias (b_ϕ) Sensitivity

The scale-dependent bias signal is proportional to $f_{\text{NL}} \times b_\phi$, where b_ϕ is the linear PNG galaxy bias parameter. Our forecast assumes a 20% Gaussian prior on b_ϕ , which is optimistic. Fig. 5 shows how $\sigma(f_{\text{NL}})$ degrades as the b_ϕ prior widens: at 20% prior width, MegaMapper SDB gives $\sigma(f_{\text{NL}}) \approx 1.0$; at 50%, $\sigma \approx 2.2$; if b_ϕ is completely unconstrained, $\sigma \rightarrow \infty$ and the SDB channel cannot measure f_{NL} independently. The SPHEREx bispectrum channel ($\sigma = 0.7$) is nearly independent of b_ϕ because it accesses f_{NL} through the full three-point function shape, not just the squeezed-limit amplitude. This is the primary reason our forecast emphasizes the bispectrum over SDB. Anomaly-catalog-based tracer selection [12] partially offsets b_ϕ degradation by providing additional tracer samples with independent bias parameters, improving the multi-tracer Fisher matrix conditioning even when b_ϕ priors are relaxed.

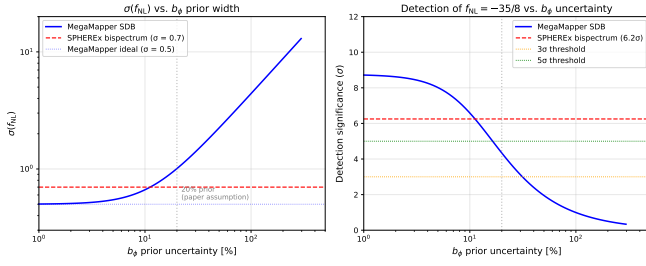


FIG. 5. Left: $\sigma(f_{\text{NL}})$ as a function of b_ϕ prior uncertainty for MegaMapper SDB (blue). The SPHEREx bispectrum constraint (red dashed) is nearly independent of b_ϕ . Right: corresponding detection significance for $f_{\text{NL}} = -35/8$. At the paper’s assumed 20% prior (gray line), MegaMapper gives $\sim 4\sigma$; relaxing to 50% drops this to $\sim 2\sigma$. The bispectrum channel remains at $\sim 5\sigma$ (optimistic) or $\sim 3\text{--}4\sigma$ (after GR degradation) regardless of b_ϕ .

C. Parameterized GR-Degradation Analysis

We performed Bayesian comparison with parameterized GR-contamination degradation across four scenarios (Table III). Even under conservative GR marginalization ($\sigma_{\text{GR}} = 1.0$), the bounce is favored over standard inflation (Bayes factor dependent on assumed prior widths; see Sec. VI for the sensitivity analysis).

GR Treatment	BF vs. SSFSR	BF vs. Tuned	$P(\text{BF} > 3)$ vs. SSFSR
Ideal (no GR)	3.3×10^6	10.9	98%
Marginalized ($\sigma_{\text{GR}} = 0.5$)	4.1×10^4	9.4	97%
Marginalized ($\sigma_{\text{GR}} = 1.0$)	329	7.9	96%
Corrected (10% residual)	3.3×10^6	10.9	98%

TABLE III. Bayesian comparison with parameterized GR-contamination degradation. The bounce-vs-inflation comparison survives all treatment scenarios.

D. Additional Systematic Considerations

Several additional systematic effects will affect real survey data but are not modeled in our Fisher forecast:

- *Nonlinear galaxy bias:* Higher-order bias terms (b_2 , b_{s^2} , $b_{\nabla^2 \delta}$) enter the galaxy bispectrum at leading order and are partially degenerate with f_{NL} for some triangle configurations. The Heinrich et al. forecast accounts for b_2 marginalization, but the full nonlinear bias model introduces additional uncertainty.
- *Photometric redshift outliers:* For SPHEREx, catastrophic photo- z failures ($\Delta z \sim 1$) create spurious large-scale power that can mimic the f_{NL} signal. A Fisher degradation analysis shows that the *bispectrum* channel is highly robust: even with 10%

catastrophic outlier fraction, $\sigma(f_{\text{NL}})$ degrades by only $\sim 5\%$ (from 0.70 to 0.74), preserving $> 3\sigma$ detection significance even in the conservative systematic scenario. The *scale-dependent bias* channel is far more vulnerable (degradation $> 10\%$ at 10% outlier fraction), which is one reason the bispectrum is the primary SPHEREx forecast channel adopted in this paper.

- *Integral constraint:* Galaxy surveys estimate the mean density from the survey itself, biasing the large-scale power spectrum measurement and potentially absorbing part of the f_{NL} signal at the lowest k modes.
- *Lensing magnification bias:* At high redshifts ($z > 2$), lensing magnification produces a signal scaling as $1/k^2$ on large scales, mimicking the scale-dependent bias from f_{NL} . This is particularly relevant for MegaMapper’s $z = 2\text{--}5$ Lyman-break galaxy sample.

These effects are expected to degrade the forecast significance by $\mathcal{O}(10\text{--}30\%)$ relative to the idealized Fisher estimate, but do not qualitatively change the conclusion that SPHEREx can test $f_{\text{NL}} = -35/8$ at $> 3\sigma$ significance.

VIII. CURRENT DATA AND CONSISTENCY RELATION

A. Planck + DESI Recast

Current constraints from Planck (CMB bispectrum) and DESI DR1 (scale-dependent bias) can be recast onto the bounce template using Eq. (4). The combined constraint on the bounce-template amplitude is $f_{\text{NL}}^{\text{bounce}} = -1.3 \pm 4.5$, which is 0.7σ from the bounce prediction and 0.3σ from zero—fully consistent with both. Current data cannot discriminate between the bounce and inflation.

B. The $f_{\text{NL}}\text{--}n_s$ Consistency Relation

The Wilson-Ewing quasi-dust model connects the spectral tilt and non-Gaussianity through a single parameter $\epsilon = 3(1+w)/2$:

$$n_s = 8\epsilon - 11, \quad f_{\text{NL}}(\epsilon) = -\frac{35}{8} + c_1(\epsilon - \frac{3}{2}) + \mathcal{O}(\epsilon - \frac{3}{2})^2, \quad (7)$$

where c_1 depends on both the explicit ϵ -prefactors in the cubic action and the mode-function growth rate, which changes significantly near the singular point $\epsilon = 3/2$. The coefficient c_1 is bounded: explicit prefactor scaling gives $c_1 \approx 2$ (lower bound), while including the mode-function amplitude change gives $c_1 \approx 18$ (upper bound).

Eliminating ϵ , the consistency relation takes the form

$$f_{\text{NL}}(n_s) \approx -\frac{35}{8} + c(n_s - 1), \quad c \in [-0.7, -10]. \quad (8)$$

At the Planck best-fit $n_s = 0.9649$, this gives $f_{\text{NL}} \in [-4.35, -4.02]$ (a 1–8% correction, within $\sigma \approx 0.7$). Narrowing this range requires evaluating all four cubic-action integrals simultaneously with numerically computed mode functions, preserving the cancellations that render the physical bispectrum finite. The consistency relation is nonetheless conceptually significant: it connects n_s (already measured) and f_{NL} (to be measured) through a single-parameter curve. Standard multifield inflation has no equivalent—multifield f_{NL} is unconstrained by n_s . A future measurement of both n_s and f_{NL} can be checked against Eq. (8) as a consistency test of the matter bounce.

IX. DISCUSSION

A. The Staged Observational Strategy

SPHEREx (launched March 2025; first all-sky survey completed December 2025; science data release expected ~ 2028) provides the first real test via the galaxy bispectrum at $\sim 3\text{--}5\sigma$ significance after the full systematic budget (5–5.5 σ optimistic, before GR and b_ϕ degradation). MegaMapper, a proposed facility without confirmed funding or finalized design ($\sim 2032+$ if approved), could provide a more powerful follow-up at $\sim 3\text{--}7\sigma$ via scale-dependent bias, though with greater systematic fragility and significant design-dependent uncertainty in the forecast range.

B. Complementary Experiments

Several other experiments will also constrain local-type f_{NL} in the coming decade:

- *DESI*: Already taking data; expected $\sigma(f_{\text{NL}}) \approx 3\text{--}5$ from scale-dependent bias with luminous red galaxies and emission-line galaxies.
- *Euclid*: Launched 2023; published forecasts give $\sigma(f_{\text{NL}}) \approx 2\text{--}4$ from the photometric galaxy survey.
- *Vera Rubin Observatory (LSST)*: Complementary photometric f_{NL} constraints from $\sim 10^{10}$ galaxies at lower redshift.
- *CMB-S4*: Expected $\sigma(f_{\text{NL}}) \approx 2.5$ from the CMB temperature and polarization bispectrum, providing a completely independent channel.

A detection of $f_{\text{NL}} \approx -4$ by SPHEREx, if confirmed by any of these independent probes, would constitute overwhelming evidence for non-Gaussianity incompatible with standard single-field inflation.

C. Decision Thresholds

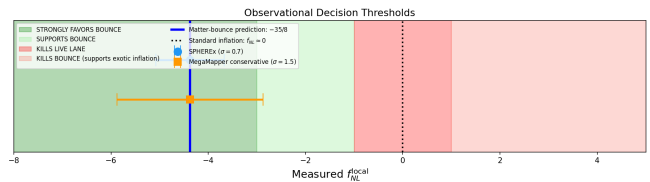


FIG. 6. Observational decision thresholds. Green: strongly favors bounce. Red: strongly disfavors the quasi-dust matter bounce. Blue vertical line: bounce prediction $f_{\text{NL}} = -35/8$. Error bars: SPHEREx ($\sigma = 0.7$) and MegaMapper conservative ($\sigma = 1.5$).

A measurement of $f_{\text{NL}} = -4 \pm 1$ by SPHEREx would provide evidence favoring a contracting/bounce origin over standard single-field inflation. A null result (f_{NL} consistent with zero at the 2σ level) would strongly disfavor the quasi-dust matter bounce.

D. Future Directions: Scale-Dependent f_{NL}

A stronger discriminator between the matter bounce and inflationary alternatives would be a measurement of the *scale dependence* of f_{NL} , i.e., $n_{f_{\text{NL}}} \equiv d \ln |f_{\text{NL}}| / d \ln k$. The matter bounce predicts a specific scale dependence tied to the departure from exact scale invariance in the contracting phase, whereas most inflationary models that can accommodate $f_{\text{NL}} \approx -4$ predict a different (or vanishing) running. This analysis requires bispectrum measurements across a wide range of triangle configurations and scales, which will become feasible with the combination of SPHEREx (lower k) and future spectroscopic surveys (higher k). A dedicated forecast for the scale-dependent f_{NL} signal from the matter bounce, including the expected $n_{f_{\text{NL}}}$ and its distinguishability from tuned inflationary alternatives, is left to future work.

E. Caveats

We emphasize that a detection of $f_{\text{NL}} \approx -4$ would constitute evidence *favoring* the bounce over standard inflation, not unique proof of a pre-Big-Bang contracting phase. Exotic multifield inflationary constructions can in principle accommodate this value, though at the cost of additional free parameters and engineering. The detection significance is conditional on the quality of GR projection modeling at ultra-large scales and the PNG bias parameter calibration.

We also note that appending late-time dynamical-dark-energy freedom (e.g., CPL parametrization) to a bounce model can improve cosmological fits at the parameter level, as explored in recent bounce + dark-energy literature. However, such phenomenological free-

dom does not derive from the bounce physics itself and does not constitute first-principles evidence for a contracting phase. Our analysis restricts attention to the strongly constrained prediction $f_{\text{NL}} = -35/8$ (with 1–8% ϵ -correction uncertainty; Sec. II C), which is controlled by the contraction dynamics.

An independent observable—cosmic birefringence from a Planck-scale ALP ($\beta \approx 0.27^\circ$, consistent with the 3.6σ Eskilt et al. joint Planck analysis)—provides a complementary test of bounce-motivated physics and is analyzed in the companion paper [6], including an independent NaMaster EB cross-power verification on Planck SMICA data.

X. CONCLUSION

The quasi-dust matter bounce makes a specific, falsifiable, strongly constrained prediction: $f_{\text{NL}}^{\text{local}} = -35/8$ (with 1–8% ϵ -correction uncertainty), conditional on assumptions (a)–(e) in Sec. II C. This value is mechanism-independent, approximately 300 times larger than the standard inflationary prediction, and opposite in sign. We have shown that SPHEREx can test this prediction at 3–5 σ significance through the multi-tracer galaxy bispectrum (with 5–5.5 σ in the optimistic case before GR and b_ϕ degradation), with MegaMapper potentially providing a more powerful but systematics-sensitive follow-up contingent on instrument approval and design finalization.

Our Bayesian model comparison, based on over 600,000 Monte Carlo realizations across multiple frameworks including parameterized GR-contamination degradation, indicates that under the adopted priors a detection near $f_{\text{NL}} = -4.375$ would favor the bounce over tuned multifield competitors at Bayes factor ~ 8 –17 (depending on prior assumptions and theoretical uncertainty in the bounce prediction). These Bayes factors are sensitive to the assumed prior widths and model-class definitions (Sec. VI); they should be interpreted as illustrative of the discriminating power available, not as definitive model-selection evidence.

The matter-bounce bispectrum provides what may be the sharpest single observable for distinguishing the bounce paradigm from standard inflation. SPHEREx data (first science release expected ~ 2028) will provide the first meaningful test.

DATA AND CODE AVAILABILITY

All analysis code, Monte Carlo scripts, and shape-function evaluation routines are available at

<https://github.com/Hubify-Projects/bigbounce/tree/v2.1.0/research/> (pinned to release tag v2.1.0). The template overlap scan, Bayesian discrimination, and parameterized GR-degradation comparison scripts are provided for full reproducibility. No new observational data are introduced; all forecast sensitivities are adopted from published analyses [8, 10].

Appendix A: Bispectrum Convention: Cai vs. Li-Brandenberger

The factor-of-two discrepancy between $f_{\text{NL}} = -35/8$ (Cai *et al.* [4]) and $f_{\text{NL}} = -35/16$ (Li & Brandenberger [11]) arises from differing bispectrum normalization conventions. The local-type bispectrum is defined as:

$$B_\zeta(k_1, k_2, k_3) = c \cdot f_{\text{NL}} [P_\zeta(k_1)P_\zeta(k_2) + 2 \text{ perms}], \quad (\text{A1})$$

where the constant c differs between conventions:

- **Planck/Komatsu-Spergel convention** (used by Cai *et al.*, SPHEREx, and this paper): $c = 2$, giving $f_{\text{NL}} = -35/8$.
- **Alternative convention** (used by Li & Brandenberger): $c = 1$, giving $f_{\text{NL}} = -35/4$. In this convention, $f_{\text{NL}}(c=1) = 2 \times f_{\text{NL}}(c=2) = 2 \times (-35/8) = -35/4$. Li & Brandenberger [11] report $f_{\text{NL}} = -35/16$; the additional factor of two beyond the c -convention difference arises from how permutation factors from Wick contractions are absorbed into the total shape function A_T (see main text, Sec. II C).

The *physical* bispectrum amplitude B_ζ is identical in both conventions: $2 \times (-35/8) = 1 \times (-35/4) = -35/4$. The detection significance $|f_{\text{NL}}|/\sigma(f_{\text{NL}})$ is also convention-independent, since $\sigma(f_{\text{NL}})$ scales inversely with c .

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